

Nmap Scan Report

Task: Nmap Network Scanning
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Scan Date

27 June 2026

Target System

Parameter	Value
Target Host	localhost
IP Address	127.0.0.1
Device Type	Local Computer
Operating System	Microsoft Windows 11 (23H2)

Commands Used

The following Nmap commands were executed during the scanning process:

```
nmap --version
nmap localhost
nmap -sV localhost
nmap -O localhost
```

Purpose of Commands

Command	Purpose
nmap --version	Verify that Nmap is installed successfully.
nmap localhost	Scan localhost to identify open ports and running services.

Command	Purpose
<code>nmap -sV localhost</code>	Detect the services running on open ports and attempt to identify their versions.
<code>nmap -O localhost</code>	Detect the operating system running on the target machine.

Open Ports

The scan identified the following open TCP ports:

Port Number	Protocol	State	Service
135	TCP	Open	MSRPC
445	TCP	Open	Microsoft-DS (SMB)

A total of **998 TCP ports** were found to be closed.

Running Services

Port	Service	Description
135	Microsoft Remote Procedure Call (MSRPC)	Enables communication between Windows applications and services.
445	Microsoft Directory Services (SMB)	Provides file and printer sharing services over the network.

During the service version detection scan, the exact software versions could not be identified because the Windows services did not expose version information.

Operating System

The operating system detection scan identified the target system as:

Parameter	Value
Device Type	General Purpose
Operating System	Microsoft Windows 11

Parameter	Value
OS Version	Windows 11 23H2
Network Distance	0 Hops

The scan correctly identified the local machine as a Windows 11 system.

Observations

- Nmap version **7.99** was successfully installed and verified.
 - The localhost responded successfully to all scan requests.
 - Two TCP ports (135 and 445) were found to be open.
 - Port 135 is associated with Microsoft Remote Procedure Call (MSRPC).
 - Port 445 is associated with SMB (Microsoft Directory Services), which enables Windows file and printer sharing.
 - The remaining 998 scanned ports were closed, reducing unnecessary exposure.
 - Service version detection was attempted, but Windows did not disclose version information.
 - Operating system detection correctly identified Microsoft Windows 11 (23H2).
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Recommendations

1. Close unnecessary open ports whenever possible.
 2. Disable SMB (Port 445) if file sharing is not required.
 3. Keep Windows and network services updated with the latest security patches.
 4. Enable Windows Firewall to restrict unauthorized network access.
 5. Perform regular vulnerability and port scans to identify new security risks.
 6. Monitor network services for suspicious activity.
 7. Remove or disable unused services to reduce the attack surface.
 8. Follow the principle of least privilege and only enable services that are required.
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Conclusion

This task provided practical experience in using Nmap for network reconnaissance and system analysis. I learned how to verify the Nmap installation, scan a target system for open ports, identify active network services, detect the operating system, and interpret scan results. The scan revealed that only two essential Windows services—MSRPC and SMB—were listening on the localhost, while the remaining scanned ports were closed. This demonstrated how Nmap can quickly provide valuable information about a system's network exposure. I also understood that open ports may represent potential security risks if they are not properly secured or are left unnecessarily enabled. Additionally, I learned that service version detection may not always reveal detailed information because operating systems can restrict version disclosure for

security reasons. Overall, this task strengthened my understanding of network scanning, reconnaissance techniques, and the importance of regularly auditing systems to identify and reduce potential security vulnerabilities. Nmap proved to be an effective tool for gathering information that supports system hardening and security assessment.